

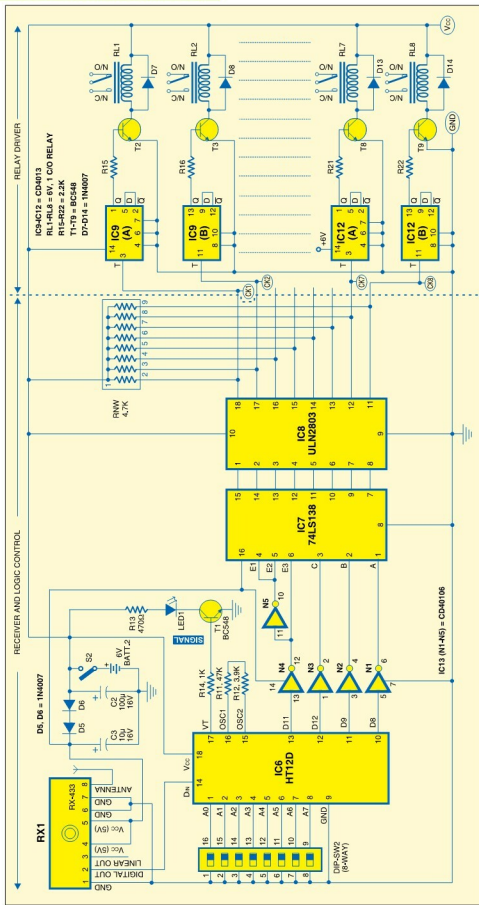


Fig. 4: Receiver circuit for PC-based wireless appliance control

to coupler (MCT2E), passes the data to the RF transmitter and the transmitter transmits the data. The RF receiver receives this data and gives it to the RF decoder (HT12D). The decoder converts the single-bit data into four-bit data and presents it to the decoder (74LS138). The output of the decoder controls the appliances with the help of flip-flops (CD4013) and transistors. Now, relays perform the corresponding action, i.e., switch the appliances 'on' or 'off'.

Remote control. For remote control, we have used the Holtek encoder-decoder pair of HT12E and HT12D. Both of these are 18-pin DIP ICs.

HT12E and HT12D are CMOS ICs with operating voltage range of 2.4V to 12V. Encoder HT12E has eight addresses and another four address/data lines. The data set on these twelve lines (address and address/data lines) is serially transmitted when the transmit-enable pin (TE) is taken low. The data output appears serially on D_{OUT} pin.

The data is transmitted four times in succession. It consists of differing lengths of positive pulses for '1' and '0', the pulse-width for '0' being twice the pulse-width for '1'. The frequency of these pulses may lie between 1.5 and 7 kHz depending on the resistor value between OSC1 and OSC2 pins.

The internal oscillator frequency of decoder HT12D is 50 times the oscillator frequency of encoder HT12E. The values of the